

DHCP — Dynamic Host Configuration Protocol

1. What is DHCP?

DHCP (Dynamic Host Configuration Protocol) is used to automatically provide network configuration information to devices.

Instead of manually configuring each device, DHCP can automatically provide:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server
- Domain Name
- Lease Time

Why do we need DHCP?

Without DHCP, every client device would need to be manually configured with an IP address and other network settings.

With DHCP:

Client Device



DHCP Server



IP Address + Subnet Mask + Gateway + DNS

DHCP is commonly used for **client devices**, such as:

- PCs
- Laptops
- Smartphones
- Tablets
- Other end-user devices

Routers and servers are usually configured with **static/manual IP addresses** because their addresses generally need to remain predictable.

DHCP in different network sizes

Small networks:

- The router commonly acts as the DHCP server.
- Example: Home Wi-Fi router.

Large networks:

- A dedicated Windows/Linux DHCP server is commonly used.
- A centralized DHCP server can serve clients across multiple subnets using a DHCP Relay Agent.

IMAGE 1 — PASTE HERE

Image: The Purpose of DHCP



- DHCP is used to allow hosts to automatically/dynamically learn various aspects of their network configuration, such as IP address, subnet mask, default gateway, DNS server, etc, without manual/static configuration.
- It is an essential part of modern networks.
→ When you connect a phone/laptop to WiFi, do you ask the network admin which IP address, subnet mask, default gateway, etc, the phone/laptop should use?
- Typically used for 'client devices' such as workstations (PCs), phones, etc.
- Devices such as routers, servers, etc, are usually manually configured.
- In small networks (such as home networks) the router typically acts as the DHCP server for hosts in the LAN.
- In larger networks, the DHCP server is usually a Windows/Linux server.

Position: Immediately after the "DHCP in different network sizes" section.

2. Basic DHCP Information

When a Windows PC receives its network configuration through DHCP, we can use:

```
ipconfig /all
```

This command displays detailed network configuration.

Example:

IPv4 Address : 192.168.0.167

Subnet Mask : 255.255.255.0
Default Gateway : 192.168.0.1
DHCP Server : 192.168.0.1
DNS Servers : 192.168.0.1

Important fields

IPv4 Address

→ IP address assigned to the PC.

Subnet Mask

→ Determines the network and host portions of the IP address.

Default Gateway

→ Router used to reach other networks.

DHCP Server

→ Device that assigned the IP configuration.

DNS Server

→ Server used to resolve domain names into IP addresses.

DHCP Enabled

If we see:

DHCP Enabled : Yes

it means the device is configured to obtain its IP configuration dynamically using DHCP.

Preferred Address

Example:

IPv4 Address : 192.168.0.167 (Preferred)

The client may request the same IP address that it previously received from the DHCP server.

This does **not** mean DHCP is giving a permanent IP address.

IMAGE 2 — PASTE HERE

Image: `ipconfig /all` showing IPv4 Address, Subnet Mask, DHCP Server, Gateway, etc.

The Basic Functions of DHCP

```
C:\Users\user>ipconfig /all

[output omitted]

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : 
    Description . . . . . : Intel(R) 82579LM Gigabit Network Connection
    Physical Address. . . . . : 78-2B-CB-AC-08-67
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 192.168.0.167(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Saturday, January 23, 2021 12:02:04 PM
    Lease Expires . . . . . : Saturday, January 23, 2021 2:02:05 PM
    Default Gateway . . . . . : 192.168.0.1
    DHCP Server . . . . . : 192.168.0.1
    DNS Servers . . . . . : 192.168.0.1
    NetBIOS over Tcpip. . . . . : Enabled

[output omitted]
```

Position: Immediately after the `ipconfig /all` explanation.

IMAGE 3 — PASTE HERE

Image: Preferred IP address explanation.

The Basic Functions of DHCP

```
C:\Users\user>ipconfig /all
```

```
[output omitted]
```

```
Ethernet adapter Ethernet0:
```

```
Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) 82579LM Gigabit Network Connection
Physical Address. . . . . : 78-2B-CB-AC-08-67
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.0.167(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, January 23, 2021 12:02:04 PM
Lease Expires . . . . . : Saturday, January 23, 2021 2:02:05 PM
Default Gateway . . . . . : 192.168.0.1
DHCP Server . . . . . : 192.168.0.1
DNS Servers . . . . . : 192.168.0.1
NetBIOS over Tcpi. . . . . : Enabled
```

This PC was previously assigned this IP address by the DHCP server, so it asked to receive the same address again this time.

```
[output omitted]
```

Position: Immediately after the "Preferred Address" section.

3. DHCP Lease

A DHCP-assigned IP address is normally given to a client for a specific period called a **lease**.

Example:

Lease Obtained : 12:02:04 PM

Lease Expires : 02:02:05 PM

This means the client has been allowed to use the IP address for a limited period.

Why use a lease?

IP addresses are limited resources.

When a device no longer needs an address, the DHCP server can eventually make that address available for another device.

Important terms

Lease Obtained

→ Time at which the client received the IP address.

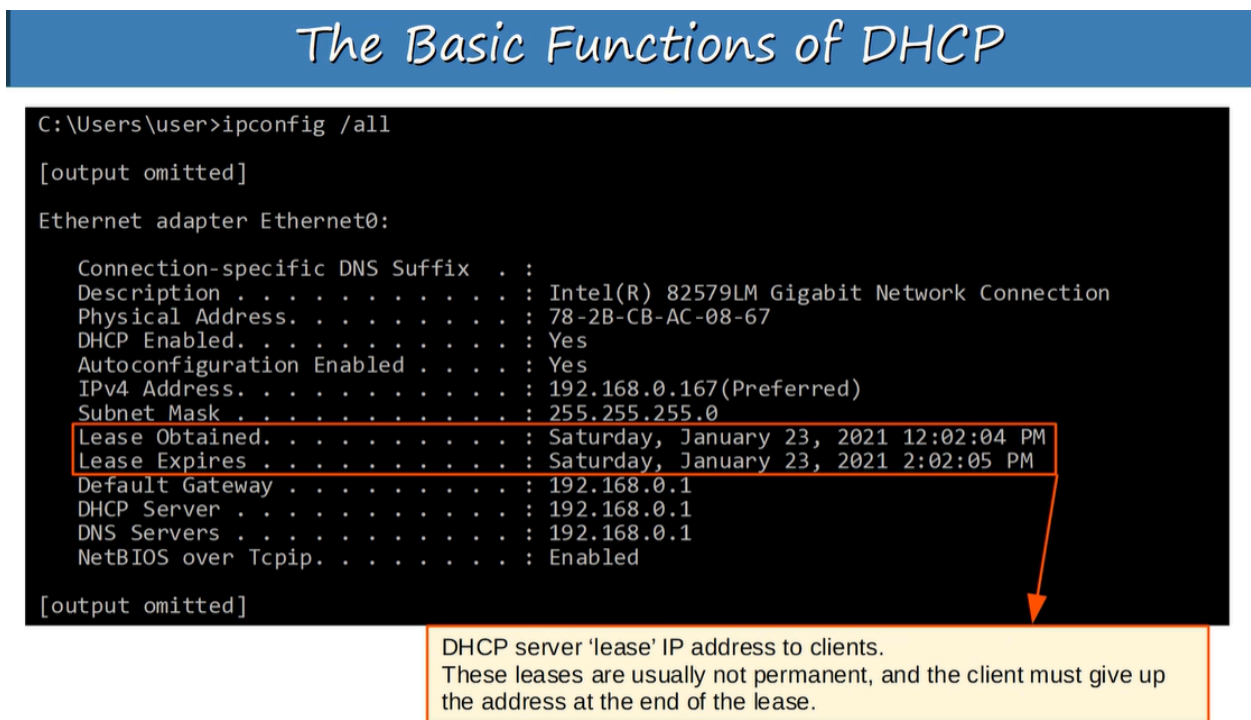
Lease Expires

→ Time at which the current lease ends.

DHCP addresses are therefore generally **temporary**, rather than permanently assigned.

IMAGE 4 — PASTE HERE

Image: Lease Obtained / Lease Expires.



Position: Immediately after the DHCP Lease explanation.

4. DHCP Release

A client can voluntarily give up its DHCP address using:

`ipconfig /release`

Example:

```
C:\Users\user>ipconfig /release
```

After the release:

Client → DHCP Server
→ "I don't need this IP address anymore."

The IP address becomes available for assignment again.

Important

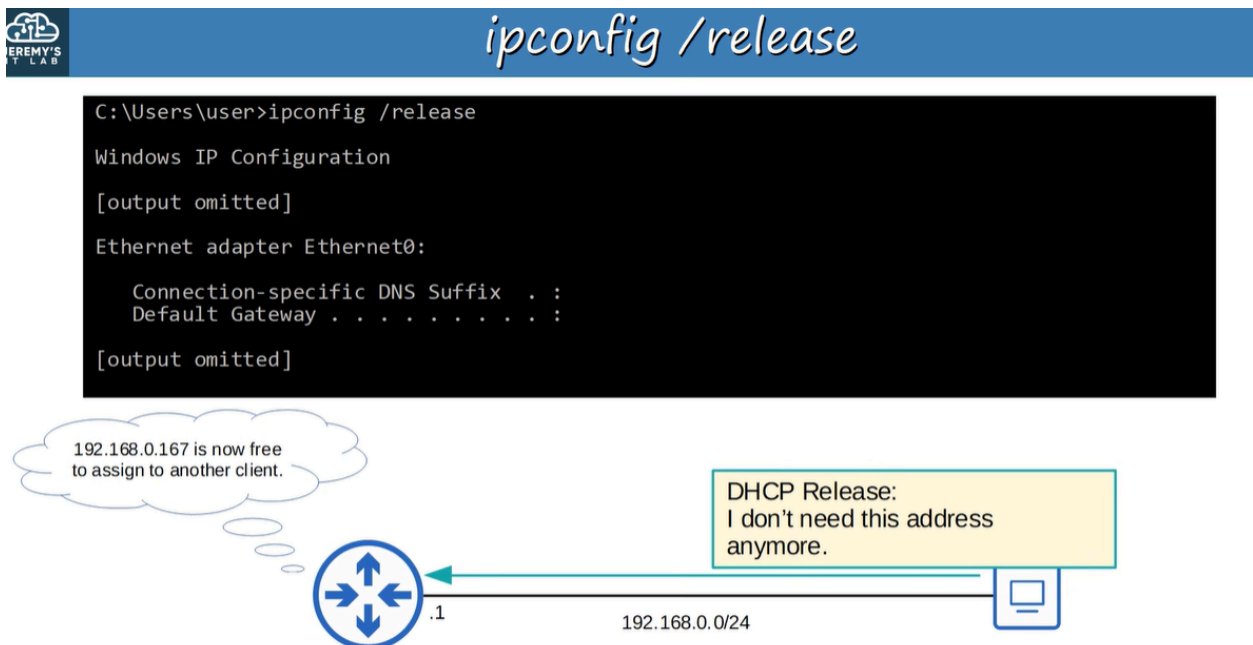
`ipconfig /release`

→ Releases the current DHCP configuration.

It does **not** request a new IP address.

📌 IMAGE 5 — PASTE HERE

Image: `ipconfig /release` command.



Position: Immediately after the DHCP Release explanation.

5. DHCP Release Packet

DHCP uses UDP.

DHCP Server → UDP Port 67

DHCP Client → UDP Port 68

For a DHCP Release:

Client

Source Port : 68

Destination Port : 67

The client sends the DHCP Release message to the DHCP server.

Example:

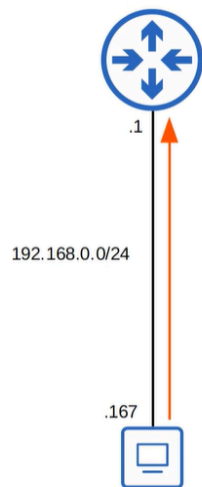
Source : 192.168.0.167

Destination : 192.168.0.1

Protocol : DHCP

 **IMAGE 6 — PASTE HERE**

Image: Wireshark DHCP Release packet showing UDP source port 68 and destination port 67.

[illegible]

DHCP **servers** use UDP **67**.
DHCP **clients** use UDP **68**.

Position: Immediately after the UDP port explanation.

6. DHCP Renew

A DHCP client can renew its lease using:

```
ipconfig /renew
```

Example:

```
C:\Users\user>ipconfig /renew
```

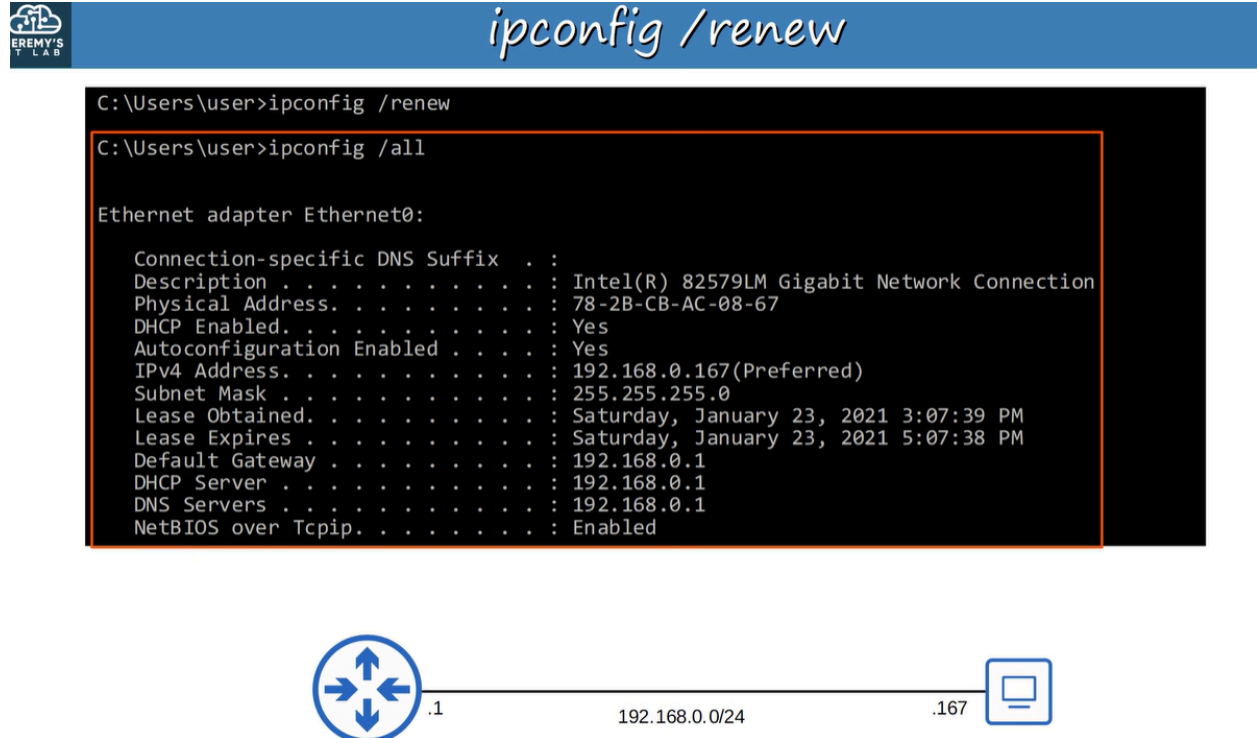
The client communicates with the DHCP server to obtain/renew its DHCP configuration.

The server can provide:

- IP address
- Subnet mask
- Default gateway
- DNS server
- New lease information

IMAGE 7 — PASTE HERE

Image: `ipconfig /renew`.



Position: Immediately after the DHCP Renew explanation.

7. DHCP Discover

When a client needs DHCP configuration, it sends a **DHCP Discover** message.

The client basically says:

"Are there any DHCP servers available? I need an IP address."

The client initially does not have a usable IP address.

Typical packet:

Source IP : 0.0.0.0

Destination IP : 255.255.255.255

Source Port : 68
Destination Port : 67

The Discover message is normally sent as a **broadcast** because the client does not yet know the DHCP server's address.

📌 IMAGE 8 — PASTE HERE

Image: Wireshark DHCP Discover packet.

The screenshot shows a Wireshark capture of a DHCP Discover packet. The packet list on the left shows a packet of type DHCP, length 342, with a transaction ID of 0xd7a1c480. The packet details on the right show the Dynamic Host Configuration Protocol (Discover) section. The fields in this section include: Message type: Boot Request (1), Hardware type: Ethernet (0x01), Hardware address length: 6, Hops: 0, Transaction ID: 0xd7a1c480, Seconds elapsed: 0, Bootp flags: 0x0000 (Unicast), Client IP address: 0.0.0.0, Your (client) IP address: 0.0.0.0, Next server IP address: 0.0.0.0, Relay agent IP address: 0.0.0.0, Client MAC address: Dell_ac:08:67 (78:2b:cb:ac:08:67), Client hardware address padding: 00000000000000000000, Server host name not given, Boot file name not given, Magic cookie: DHCP, Option (53) DHCP Message Type (Discover), Option (61) Client identifier, Option (50) Requested IP Address (192.168.0.167), Option (12) Host Name, Option (60) Vendor class identifier, Option (55) Parameter Request List, Option (255) End, and Padding: 000000000000000000000000.

Position: Immediately after the DHCP Discover explanation.

Note: The screenshot title says "DHCP Release", but the packet shown is a **DHCP Discover**. Use the packet information itself for revision.

8. DHCP Offer

The DHCP server responds with a **DHCP Offer**.

The server basically says:

"How about this IP address?"

Example:

Offered IP Address : 192.168.0.167
Subnet Mask : 255.255.255.0
Gateway : 192.168.0.1
DNS Server : 192.168.0.1

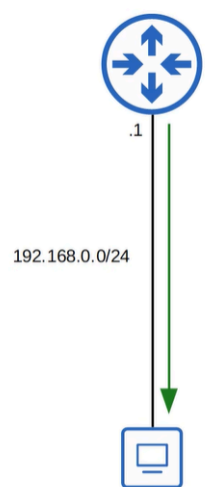
The DHCP Offer can be:

- Broadcast
- Unicast

depending on the communication stage and client capabilities.

IMAGE 9 — PASTE HERE

Image: DHCP Offer Wireshark packet.



DHCP Offer

No.	Time	Source	Destination	Protocol	Length	Info
262	13:27:34.562795	192.168.0.1	192.168.0.167	DHCP	342	DHCP Offer - Transaction ID 0xd7a1c480

> Frame 262: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface \Device\NPF_{9956EC07-3774-4B11-970C-...}

> Ethernet II, Src: Tp-LinkT_dd:a8:e4 (98:da:c4:dd:a8:e4), Dst: Dell_ac:08:67 (78:2b:cb:ac:08:67)

> Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.167

> User Datagram Protocol, Src Port: 67, Dst Port: 68

> Dynamic Host Configuration Protocol (Offer)

Message type: Boot Reply (2)
Hardware type: Ethernet (0x01)
Hardware address length: 6
Hops: 0
Transaction ID: 0xd7a1c480
Seconds elapsed: 0
Bootp flags: 0x0000 (Unicast)
Client IP address: 0.0.0.0
Your (client) IP address: 192.168.0.167
Next server IP address: 192.168.0.1
Relay agent IP address: 0.0.0.0
Client MAC address: Dell_ac:08:67 (78:2b:cb:ac:08:67)
Client hardware address padding: 00000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP

> Option: (53) DHCP Message Type (Offer)
> Option: (54) DHCP Server Identifier (192.168.0.1)
> Option: (51) IP Address Lease Time
> Option: (58) Renewal Time Value
> Option: (59) Rebinding Time Value
> Option: (1) Subnet Mask (255.255.255.0)
> Option: (28) Broadcast Address (192.168.0.255)
> Option: (6) Domain Name Server
> Option: (3) Router
> Option: (255) End
Padding: 0000000000000000

The DHCP Offer message can be either **broadcast** or **unicast**.

Position: Immediately after the DHCP Offer explanation.

9. DHCP Request

The client receives one or more DHCP Offers.

The client then sends a **DHCP Request**.

The client basically says:

"I want to use the IP address you offered me."

Example:

Requested IP Address : 192.168.0.167

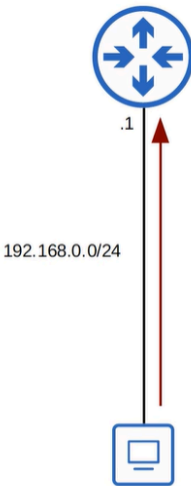
DHCP Server Identifier: 192.168.0.1

The DHCP Request is normally sent as a **broadcast** during the initial DORA process.

This also allows other DHCP servers to know that their offers were not selected.

IMAGE 10 — PASTE HERE

Image: DHCP Request Wireshark packet.



DHCP Request

No.	Time	Source	Destination	Protocol	Length	Info
263	13:27:34.563458	0.0.0.0	255.255.255.255	DHCP	344	DHCP Request - Transaction ID 0xd7a1c480

```
> Frame 263: 344 bytes on wire (2752 bits), 344 bytes captured (2752 bits) on interface \Device\NPF_{9956EC07-3774-4811-9700}
> Ethernet II, Src: Dell_ac:08:67 (78:2b:cb:ac:08:67), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
> User Datagram Protocol, Src Port: 68, Dst Port: 67
> Dynamic Host Configuration Protocol (Request)
  Message type: Boot Request (1)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0xd7a1c480
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: Dell_ac:08:67 (78:2b:cb:ac:08:67)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  > Option: (53) DHCP Message Type (Request)
  > Option: (61) Client identifier
  > Option: (50) Requested IP Address (192.168.0.167)
  > Option: (54) DHCP Server Identifier (192.168.0.1)
  > Option: (12) Host Name
  > Option: (81) Client Fully Qualified Domain Name
  > Option: (60) Vendor class identifier
  > Option: (55) Parameter Request List
  > Option: (255) End
```



Position: Immediately after the DHCP Request explanation.

10. DHCP ACK

The DHCP server sends a **DHCP ACK (Acknowledgment)**.

The server basically says:

"Okay, you may use it."

The ACK confirms the client's DHCP configuration.

It can contain:

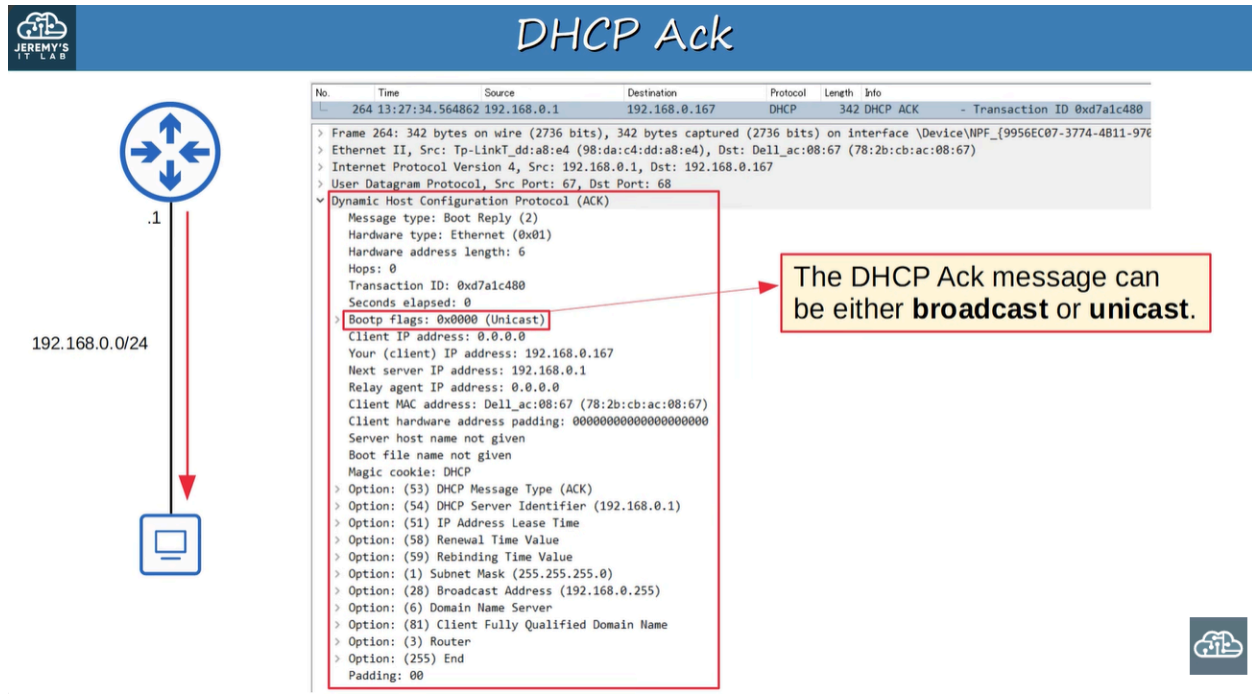
- IP address
- Subnet mask
- Default gateway
- DNS server
- Lease time
- Renewal information
- Other DHCP options

The DHCP ACK can be:

- Broadcast
 - Unicast
-

 **IMAGE 11 — PASTE HERE**

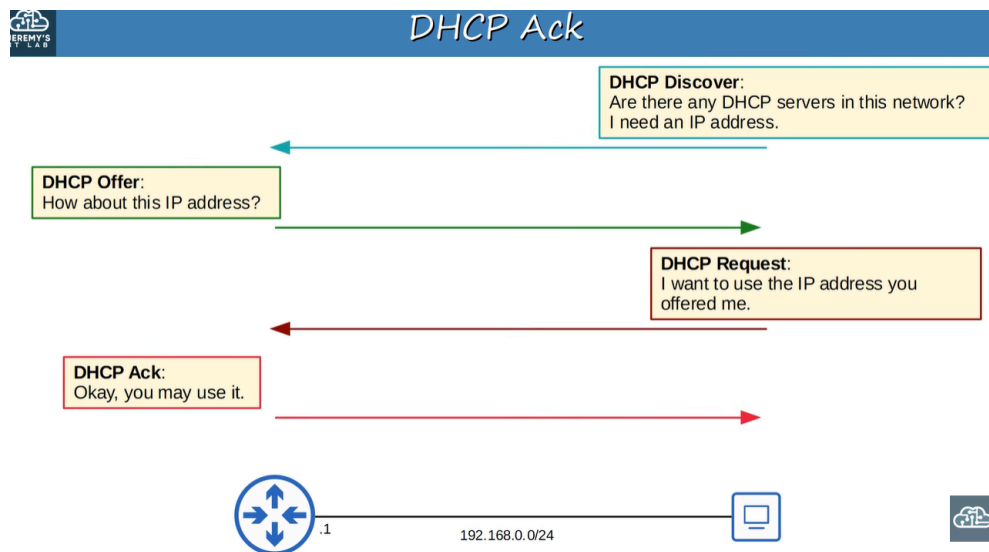
Image: DHCP ACK Wireshark packet.



The DHCP Ack message can be either **broadcast** or **unicast**.

Position: Immediately after the DHCP ACK explanation.

11. DHCP DORA Process



D-O-R-A

D = Discover

O = Offer

R = Request

A = ACK

1. Discover

Client → Server

The client searches for DHCP servers.

Usually:

Broadcast

2. Offer

Server → Client

The server offers an IP address.

Can be:

Broadcast or Unicast

3. Request

Client → Server

The client requests the offered IP address.

Usually:

Broadcast

4. ACK

Server → Client

The server confirms the IP configuration.

Can be:

Broadcast or Unicast

DORA Summary

Message	Direction	Typical Delivery
Discover	Client → Server	Broadcast
Offer	Server → Client	Broadcast / Unicast
Request	Client → Server	Broadcast
ACK	Server → Client	Broadcast / Unicast

 **IMAGE 12 — PASTE HERE**

Image: DHCP D-O-R-A table.

DHCP D-O-R-A		
Discover	Client → Server	Broadcast
Offer	Server → Client	Broadcast or Unicast
Request	Client → Server	Broadcast
Ack	Server → Client	Broadcast or Unicast
Release	Client → Server	Unicast

Position: Immediately after the DORA summary table.

12. DHCP Release vs DORA

Do not confuse **DHCP Release** with the DORA process.

DORA

Used when a client needs/obtains DHCP configuration:

Discover



Offer



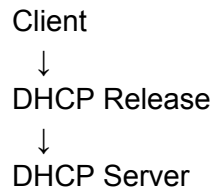
Request



ACK

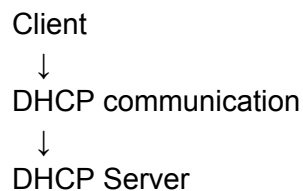
Release

Used when the client voluntarily gives up its DHCP address:



Renew

Used when the client wants to continue using DHCP configuration / renew its lease:



13. DHCP Relay

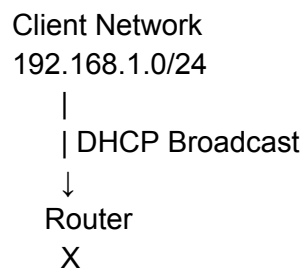
Why do we need DHCP Relay?

DHCP Discover is normally a **broadcast**.

Routers normally **do not forward broadcasts between different networks**.

Therefore, if the DHCP server is located on another subnet, the client's DHCP Discover cannot directly reach the server.

Example:



|
X Broadcast does not cross router
|
DHCP Server
192.168.10.0/24

To solve this problem, the router can act as a:

DHCP Relay Agent

The router receives the client's DHCP broadcast and forwards it toward the remote DHCP server, typically as a unicast message.

IMAGE 13 — PASTE HERE

Image: DHCP Relay explanation showing centralized DHCP server.

Position: Immediately after explaining why DHCP Relay is required.

14. DHCP Relay Example

Consider:

DHCP Server
192.168.10.10
|
|
R2
|
10.0.0.0/30
|
R1
|
PC1
192.168.1.100

Here:

DHCP Server = 192.168.10.10
PC Network = 192.168.1.0/24

The DHCP server and PC are on **different networks**.

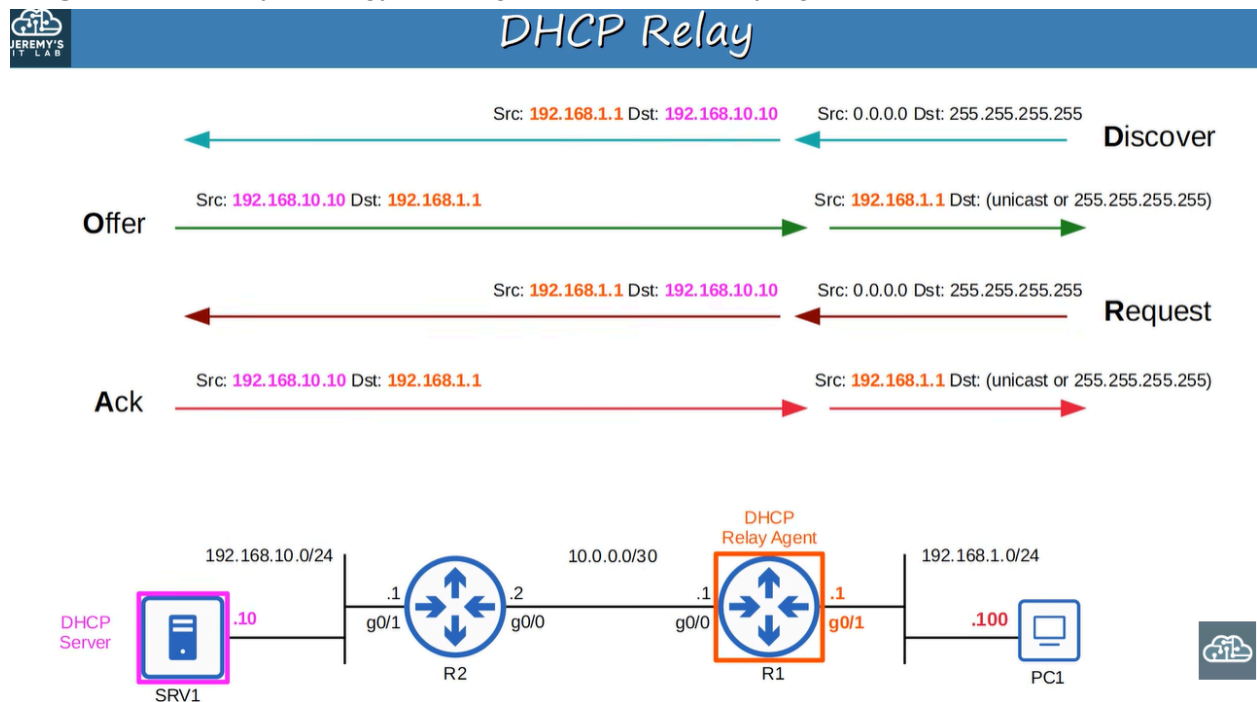
Therefore, R1 acts as the **DHCP Relay Agent**.

Important point

The DHCP relay is configured on the **router interface connected to the client subnet**.

📌 IMAGE 14 — PASTE HERE

Image: DHCP Relay topology showing R1 as DHCP Relay Agent.



Position: Immediately after the DHCP Relay Example.

15. DHCP Relay Message Flow

Without DHCP Relay:

PC
↓
DHCP Discover Broadcast
↓
Router
X
DHCP Server

The broadcast cannot cross the router.

With DHCP Relay:

PC
↓
DHCP Discover
↓
R1 — DHCP Relay Agent
↓
Unicast
↓
R2
↓
DHCP Server

The relay agent forwards the client's DHCP message to the remote DHCP server.

Message flow

Discover:

PC → Relay → DHCP Server

Offer:

DHCP Server → Relay → PC

Request:

PC → Relay → DHCP Server

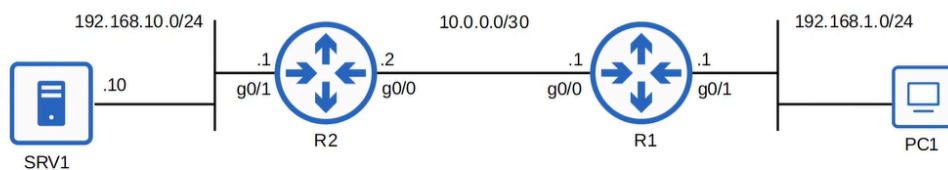
ACK:

DHCP Server → Relay → PC

Image: DHCP Relay packet-flow diagram showing source and destination IP addresses.



- Some network engineers might choose to configure each router to act as the DHCP server for its connected LANs.
- However, large enterprises often choose to use a centralized DHCP server.
- If the server is centralized, it won't receive the DHCP clients' broadcast DHCP messages. (broadcast messages don't leave the local subnet)
- To fix this, you can configure a router to act as a **DHCP relay agent**.
- The router will forward the clients' broadcast DHCP messages to the remote DHCP server as unicast messages.



Position: Immediately after the message-flow explanation.

16. DHCP Server Configuration on Cisco IOS

A Cisco router can act as a DHCP server.

Step 1 — Exclude addresses

```
R1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

These addresses will **not** be assigned to DHCP clients.

Usually we exclude addresses used by:

- Router interfaces
- Servers
- Printers
- Other devices requiring static IPs

Step 2 — Create DHCP Pool

```
R1(config)# ip dhcp pool LAB_POOL
```

Creates a DHCP pool named:

LAB_POOL

Step 3 — Specify the network

```
R1(dhcp-config)# network 192.168.1.0 /24
```

or:

```
R1(dhcp-config)# network 192.168.1.0 255.255.255.0
```

Specifies the network from which DHCP addresses will be assigned.

Step 4 — Configure DNS Server

```
R1(dhcp-config)# dns-server 8.8.8.8
```

Specifies the DNS server given to DHCP clients.

Step 5 — Configure Domain Name

```
R1(dhcp-config)# domain-name jeremysitlab.com
```

Specifies the domain name for DHCP clients.

Step 6 — Configure Default Gateway

```
R1(dhcp-config)# default-router 192.168.1.1
```

Specifies the default gateway given to clients.

Step 7 — Configure Lease Time

```
R1(dhcp-config)# lease 0 5 30
```

Format:

lease days hours minutes

Example:

```
lease 0 5 30
```

means:

0 days

5 hours

30 minutes

You can also configure:

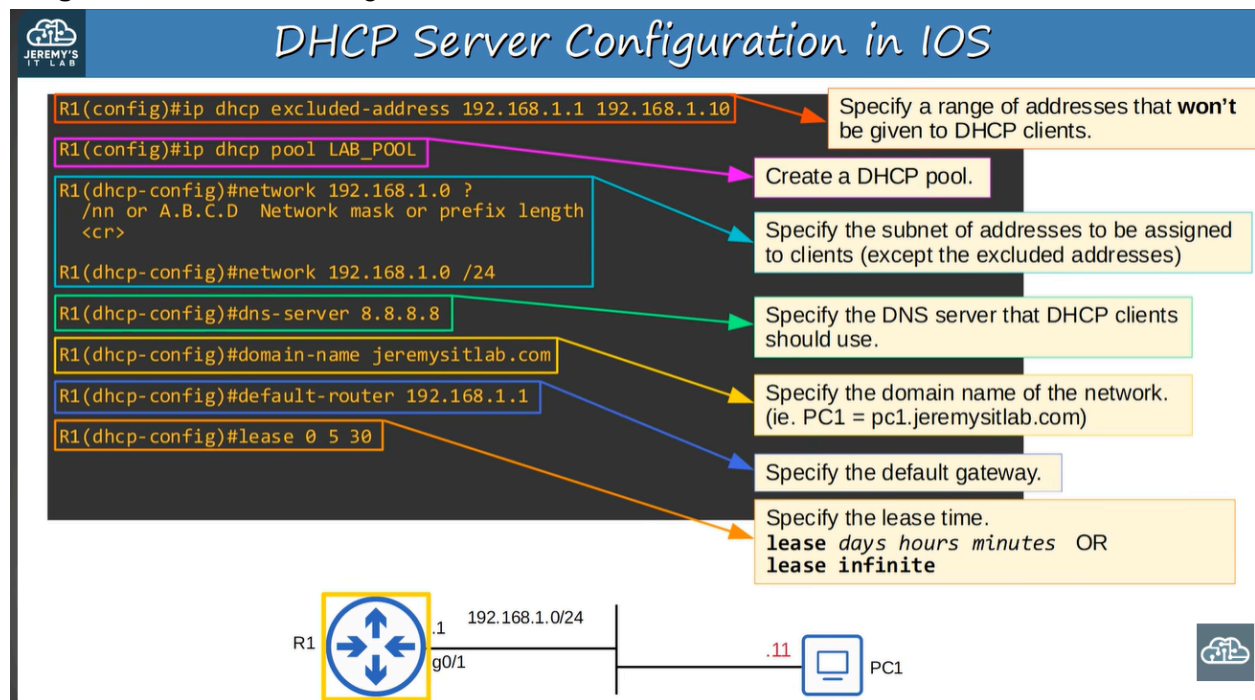
```
lease infinite
```

for an infinite lease.



IMAGE 16 — PASTE HERE

Image: DHCP Server Configuration in IOS.



Position: Immediately after the complete DHCP server configuration.

17. Verify DHCP Bindings

Use:

R1# show ip dhcp binding

This displays the IP addresses assigned by the DHCP server.

Example:

IP address	Client-ID	Lease expiration
192.168.1.11	0100.0c29...	Jan 24 2021

This helps us verify:

- Which IP was assigned
- Which client received it
- Lease expiration

- Assignment type

📌 IMAGE 17 — PASTE HERE

Image: `show ip dhcp binding` output.

DHCP Server Configuration in IOS

```
R1#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.1.11    0100.0c29.e727.39  Jan 24 2021 10:52 AM    Automatic
```

```
C:\Users\user>ipconfig /all

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : jeremysitlab.com
    Description . . . . . : Intel(R) PRO/1000 MT Network Connection #2
    Physical Address. . . . . : 00-0C-29-E7-27-39
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 192.168.1.11(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Saturday, January 24, 2021 2:22:35 PM
    Lease Expires . . . . . : Saturday, January 24, 2021 7:52:35 PM
    Default Gateway . . . . . : 192.168.1.1
    DHCP Server . . . . . : 192.168.1.1
    DNS Servers . . . . . : 8.8.8.8
    NetBIOS over Tcpip. . . . . : Enabled
```

```
graph LR
    R1((R1)) --- |192.168.1.0/24| PC1[PC1]
    style R1 fill:#fff,stroke:#000,stroke-width:2px
    style PC1 fill:#fff,stroke:#000,stroke-width:2px
```

Position: Immediately after the verification command.

18. DHCP Relay Agent Configuration

To configure a Cisco router as a DHCP Relay Agent:

```
R1(config)# interface g0/1
R1(config-if)# ip helper-address 192.168.10.10
```

Important

The interface should be the interface:

connected to the client subnet.

Example:

```
PC1
192.168.1.100
|
|
R1 G0/1
192.168.1.1
```

Configure:

```
R1(config)# interface g0/1
R1(config-if)# ip helper-address 192.168.10.10
```

Here:

192.168.10.10

is the DHCP server's IP address.

Easy memory trick

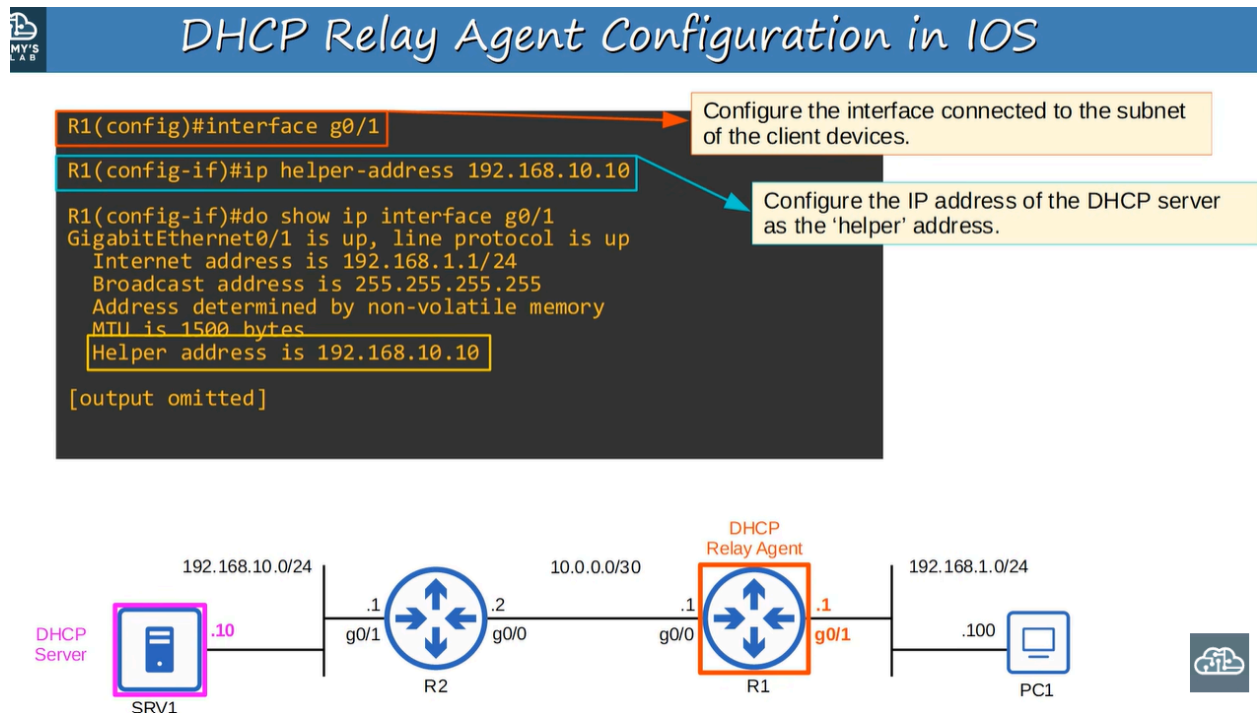
ip helper-address DHCP_SERVER_IP

means:

"Forward DHCP requests received on this interface to this DHCP server."

 **IMAGE 18 — PASTE HERE**

Image: DHCP Relay Agent Configuration in IOS.



Position: Immediately after the `ip helper-address` explanation.

19. DHCP Client Configuration on Cisco IOS

A router interface can also act as a DHCP client.

Use:

```
R2(config)# interface g0/1
R2(config-if)# ip address dhcp
```

This tells the router:

"Obtain the IP address for this interface using DHCP."

The router will automatically receive its IP configuration from a DHCP server.

Example:

```
R2(config-if)# ip address dhcp
```

Then verify using:

```
R2# show ip interface g0/1
```

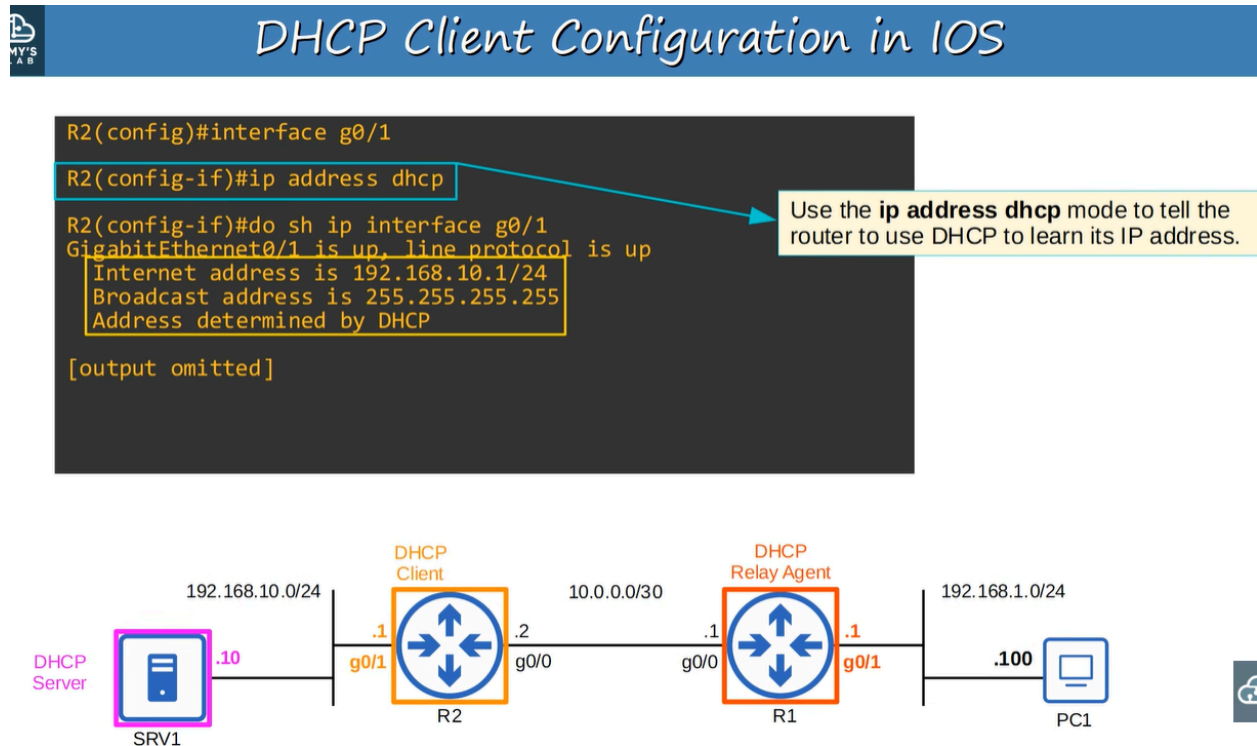
The output may show:

Internet address is 192.168.10.1/24

Address determined by DHCP

IMAGE 19 — PASTE HERE

Image: DHCP Client Configuration in IOS.



Position: Immediately after the DHCP client configuration section.

20. Important DHCP Commands — Quick Revision

Windows Client Commands

View DHCP information

```
ipconfig /all
```

Release DHCP address

```
ipconfig /release
```

Renew DHCP address

```
ipconfig /renew
```

Cisco DHCP Server Commands

Exclude addresses

```
ip dhcp excluded-address low-address high-address
```

Create DHCP pool

```
ip dhcp pool pool-name
```

Configure network

```
network ip-address { /prefix-length | subnet-mask }
```

Configure DNS

```
dns-server ip-address
```

Configure domain name

domain-name domain-name

Configure default gateway

default-router ip-address

Configure lease

lease days hours minutes

or:

lease infinite

Check DHCP bindings

show ip dhcp binding

Cisco DHCP Relay Agent

Configure on the interface connected to the **client network**:

```
interface g0/1
ip helper-address DHCP_SERVER_IP
```

Example:

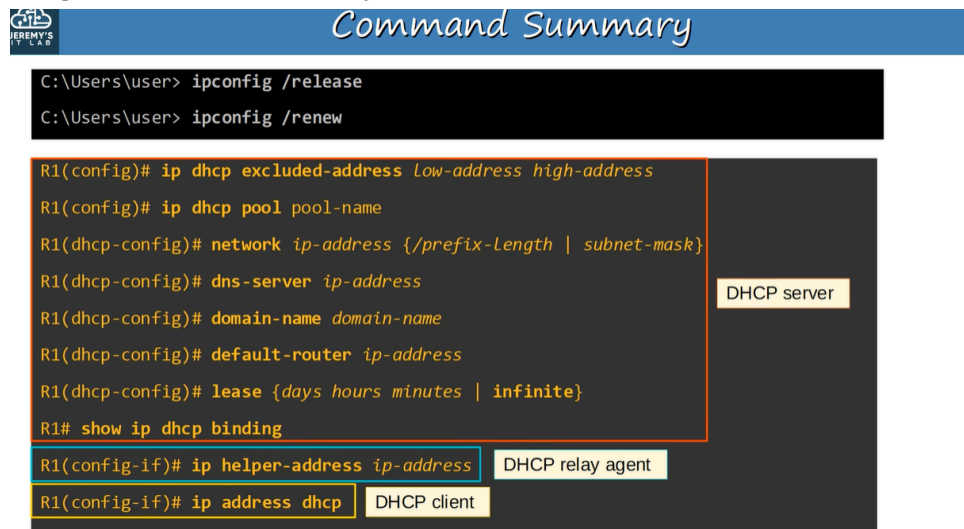
```
interface g0/1
ip helper-address 192.168.10.10
```

Cisco DHCP Client

```
interface g0/1
ip address dhcp
```

IMAGE 20 — PASTE HERE

Image: Command Summary.



The image shows a terminal window titled "Command Summary" with a blue header. It contains a list of DHCP-related commands categorized into three groups:

- Client commands:**
 - `C:\Users\user> ipconfig /release`
 - `C:\Users\user> ipconfig /renew`
- Router configuration commands (R1):**
 - `R1(config)# ip dhcp excluded-address low-address high-address`
 - `R1(config)# ip dhcp pool pool-name`
 - `R1(dhcp-config)# network ip-address {/prefix-length | subnet-mask}`
 - `R1(dhcp-config)# dns-server ip-address`
 - `R1(dhcp-config)# domain-name domain-name`
 - `R1(dhcp-config)# default-router ip-address`
 - `R1(dhcp-config)# lease {days hours minutes | infinite}`
 - `R1# show ip dhcp binding`
- Relay agent configuration:**
 - `R1(config-if)# ip helper-address ip-address` (labeled "DHCP relay agent")
 - `R1(config-if)# ip address dhcp` (labeled "DHCP client")

Position: At the very end of the DHCP notes.

21. DHCP Ports — Must Remember

DHCP uses **UDP**.

DHCP Server → UDP 67

DHCP Client → UDP 68

Easy memory:

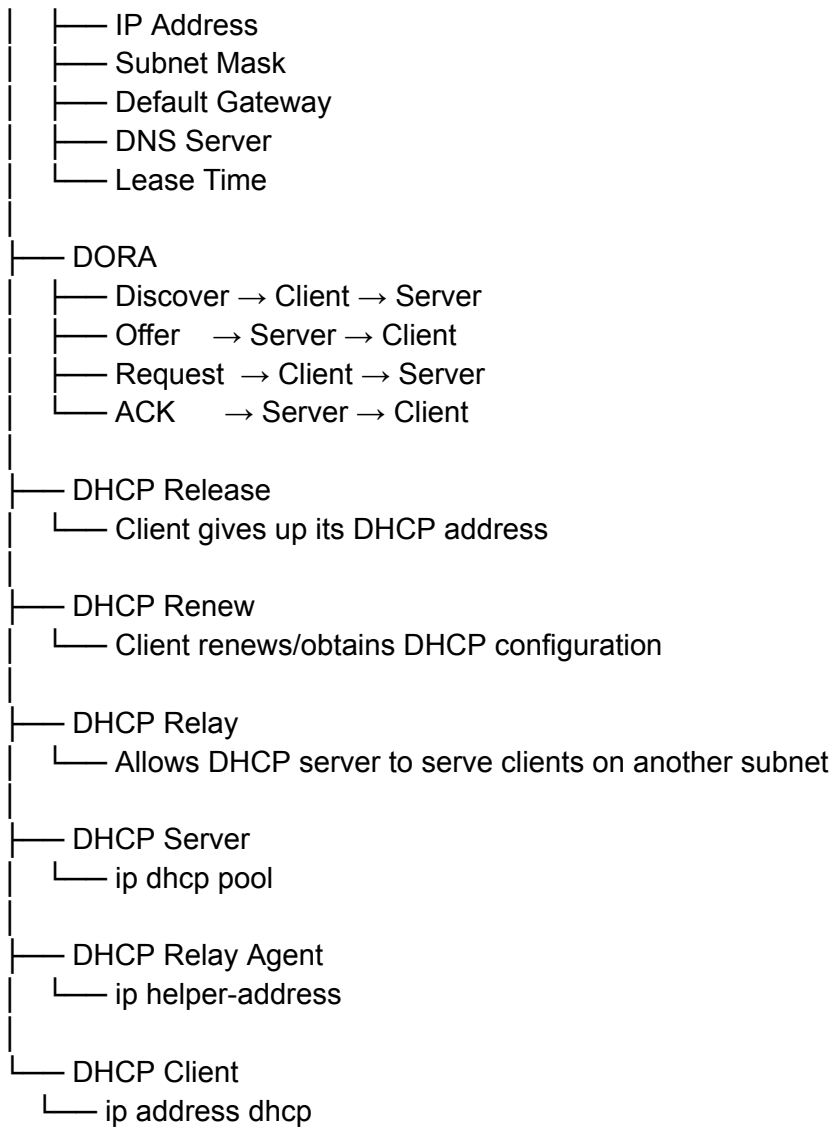
Server = 67

Client = 68

22. Final DHCP Revision

DHCP

|
| — Automatically provides network configuration



★ Exam Points

DHCP = Dynamic Host Configuration Protocol

DORA = Discover → Offer → Request → ACK

DHCP Server Port = UDP 67

DHCP Client Port = UDP 68

DHCP Broadcast does not normally cross a router.

DHCP Relay allows DHCP broadcasts to reach a remote DHCP server.

Cisco DHCP Relay command:
ip helper-address <DHCP-server-IP>

Cisco DHCP Server:
ip dhcp pool <pool-name>

Cisco DHCP Client:
ip address dhcp

Windows:
ipconfig /release
ipconfig /renew
ipconfig /all